



Seismic Isolation Bearing Design

003 Shear Stiffness

Example 04 - rivtbook

proj. 0004



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0.3i | Rubber Elastic Properties

Table 1: IRHD hardness scale (MN/m²)

Hardness IRHD +/-2	Young's modulus E, MN/m ²	Shear modulus G, MN/m ²	k	Bulk modulus B(inf), MN/m ²
30	0.92	0.30	0.93	1000
35	1.18	0.37	0.89	1000
40	1.50	0.45	0.85	1000
45	1.80	0.54	0.80	1000
50	2.20	0.64	0.73	1030
55	3.25	0.81	0.64	1090
60	4.45	1.06	0.57	1150
65	5.85	1.37	0.54	1210
70	7.35	1.73	0.53	1270
75	9.40	2.22	0.52	1330

Table 2: Shore A hardness scale (lbf/in²)

Shore A (approx.)	Young's modulus lbf/in ²	Shear modulus lbf/in ²	k	Bulk modulus lbf/in ²
35	168	53	0.89	142 000
45	256	76	0.80	142 000
55	460	115	0.64	154 000
65	830	195	0.54	171 000
75	1340	317	0.52	189 000

Note

1. The majority of springs are in the hardness range 40-60 IRHD.



2. Theoretically, with a Poisson’s ratio of 1/2, B should equal 3G. This is true for soft gum rubbers. For harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase 3G to about 4G.

0.3 - 2v | Shear Stiffness Example

Table 3: Bearing Specifications

variable	value	[value]	description
G_1	58.00 p_si	0.40 MPA	shear modulus
K_1	200000.00 p_si	1378.95 MPA	bulk modulus
rn1	51	51	number of rubber layers
rdia	39.50 inch	100.33 cm	bearing diameter
rthk	0.40 inch	1.02 cm	layer thickness
sdia	38.00 inch	96.52 cm	shim diameter
sthk	0.09 inch	0.23 cm	11 guage shim thickness
cpi	3.1418	3.1418	contant pi

Bearing Stiffness for Circular Bearing

Eq. 1: rubber height

$$rht = rn1 \cdot rthk$$
$$rht = 20.40 \text{ inch} \quad [rht] = 51.82 \text{ cm} \quad | \text{ rubber height}$$

$rthk$	$rn1$
0.40 inch	51
layer thickness	number of rubber layers

Eq. 2: bearing height

$$bht = rht + sthk \cdot (rn1 - 1)$$



bht = 24.95 inch [bht] = 63.37 cm | bearing height

rn1	sthk	rht
51	0.09 inch	20.40 inch
number of rubber layers	11 guage shim thickness	rubber height -

Eq. 3: shear stiffness

$$K_{s1} = \frac{G_1 \cdot cpi \cdot rdia^2}{4 \cdot rht}$$

$K_{s1} = 3.48 \text{ k_in}$ [K_{s1}] = 6.10 kN_cm | shear stiffness

rht	rdia	cpi	G ₁
20.40 inch	39.50 inch	3.1418	58.00 p_si
rubber height	bearing diameter	contant pi	shear modulus

Shape Factor for Circular Bearing

Eq. 4: shape factor 1

$$sh_{11} = \frac{sdia}{4 \cdot rthk}$$

$sh_{11} = 24$ [sh_{11}] = 24 | shape factor 1

rthk	sdia
0 inch	38 inch
layer thickness	shim diameter

Eq. 5: shape factor 2

$$sh_{21} = \frac{bht}{rdia}$$



sh _{2 1} = 1 [sh _{2 1}] = 1 shape factor 2	
rdia	bht
40 inch	25 inch
bearing diameter	bearing height

